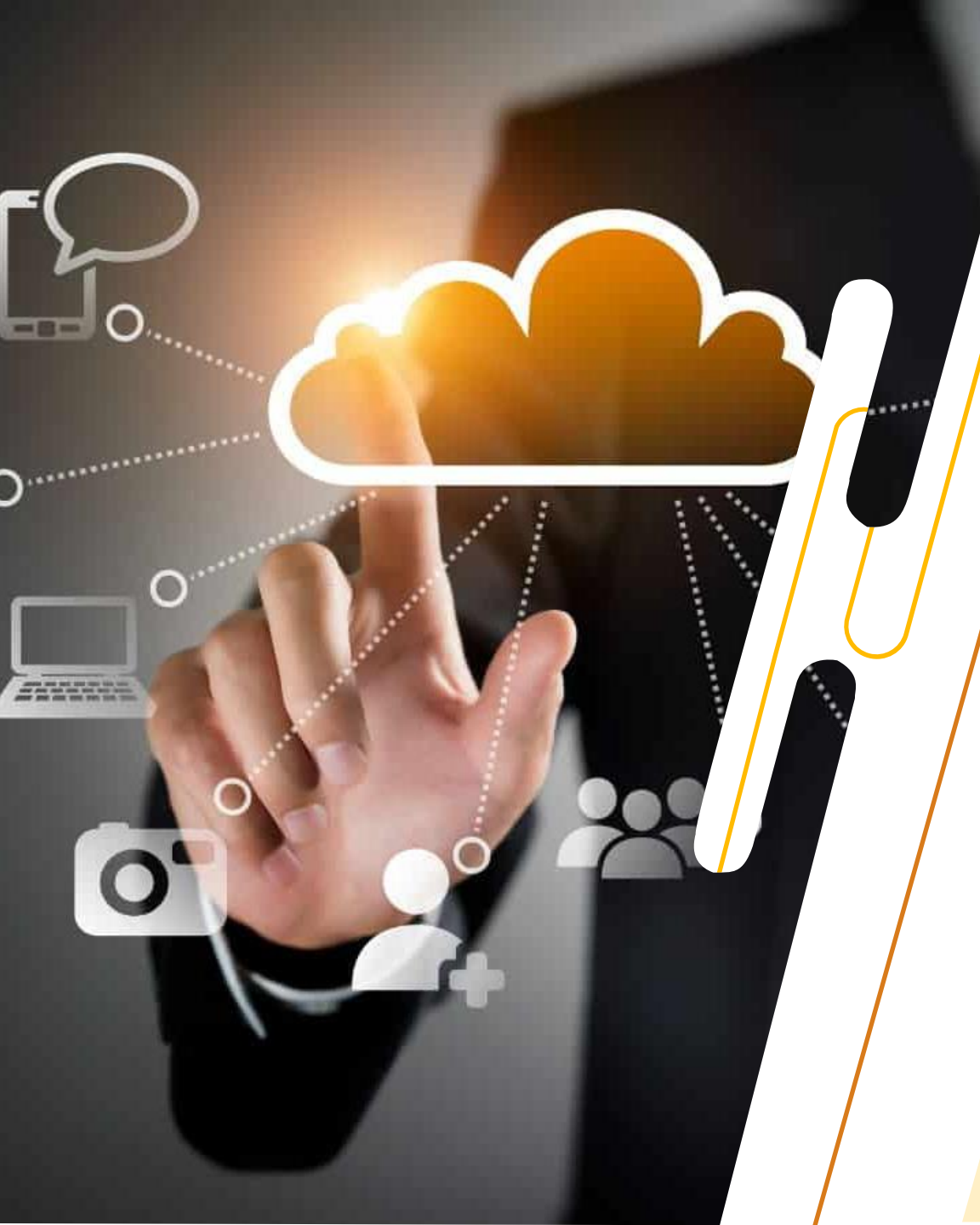




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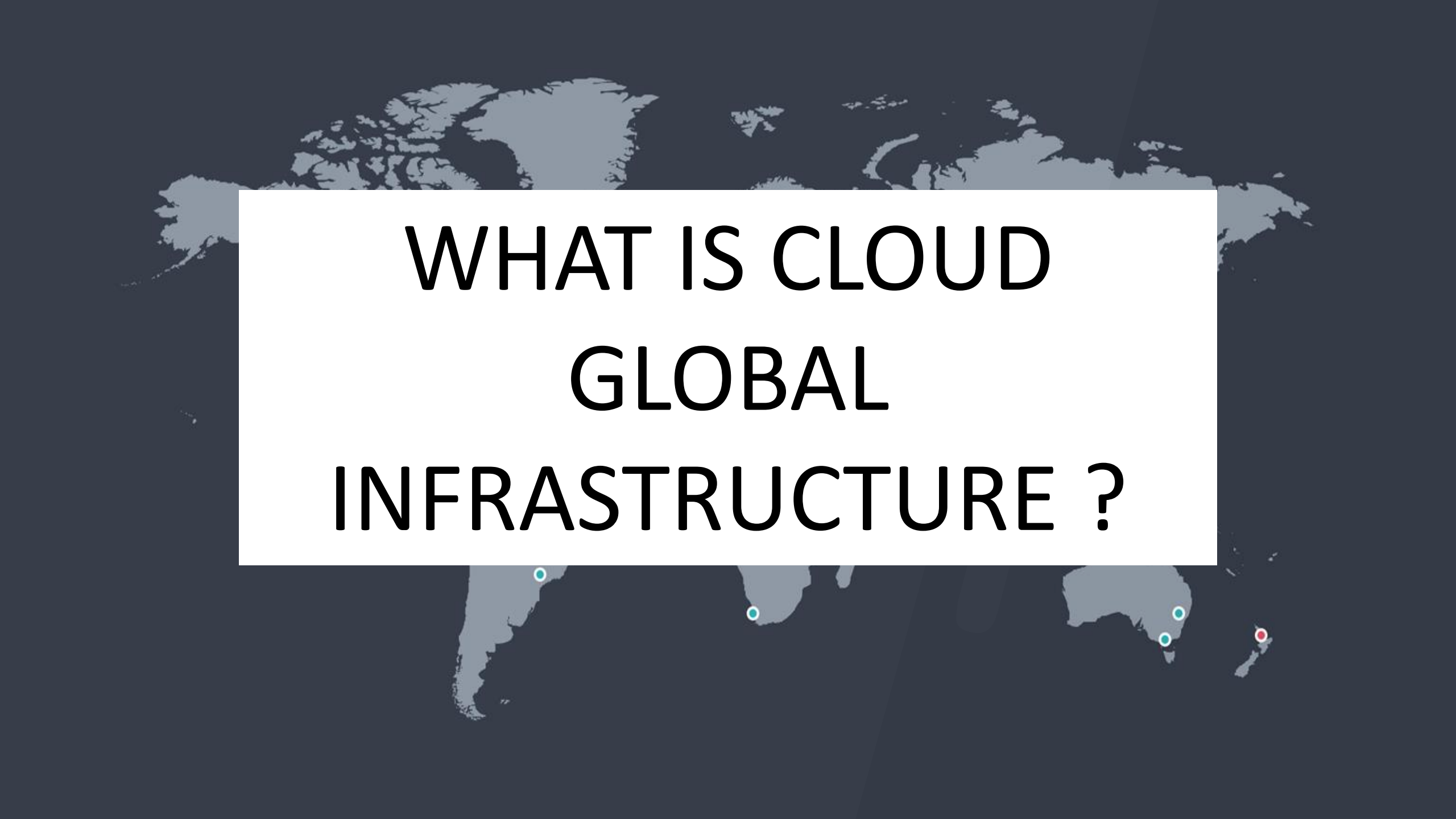
Massive Data Processing and Cloud Computing

YEAR 3 SEMESTER 2
DATA SCIENCE SPECIALIZATION

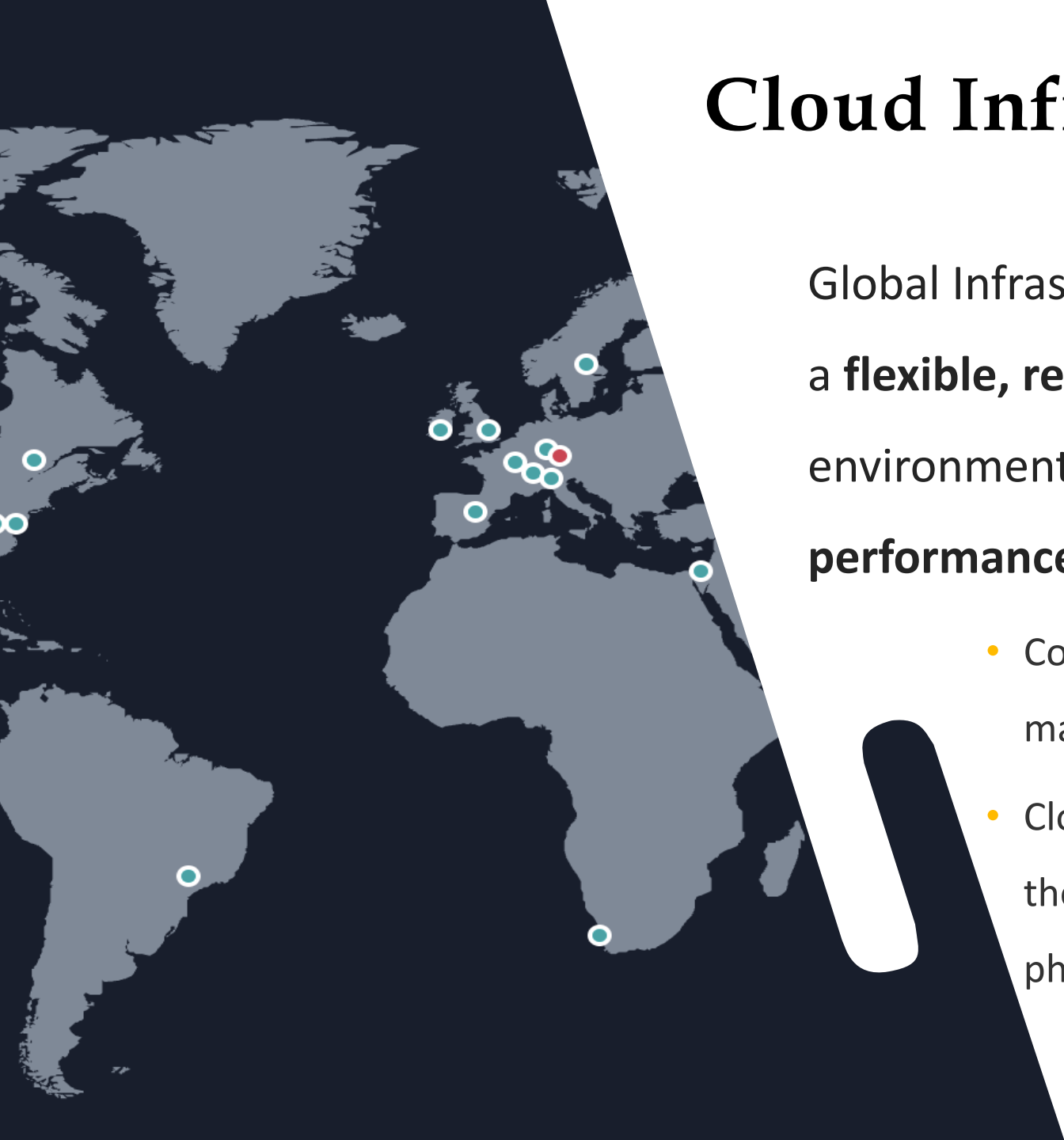
Lecture Outline

- 1. What is Cloud Global Infrastructure?
- 2. Benefits of the Global Infrastructure
- 3. Cloud Core Services
- 4. Cloud Economics and Billing



A dark blue world map serves as the background. A white rectangular box is centered over the map, containing the title text. Several small circular location pins are visible on the map: one in South America, one in Africa, one in Australia, and one in Japan. The text is in a large, bold, black, sans-serif font.

WHAT IS CLOUD GLOBAL INFRASTRUCTURE ?



Cloud Infrastructure

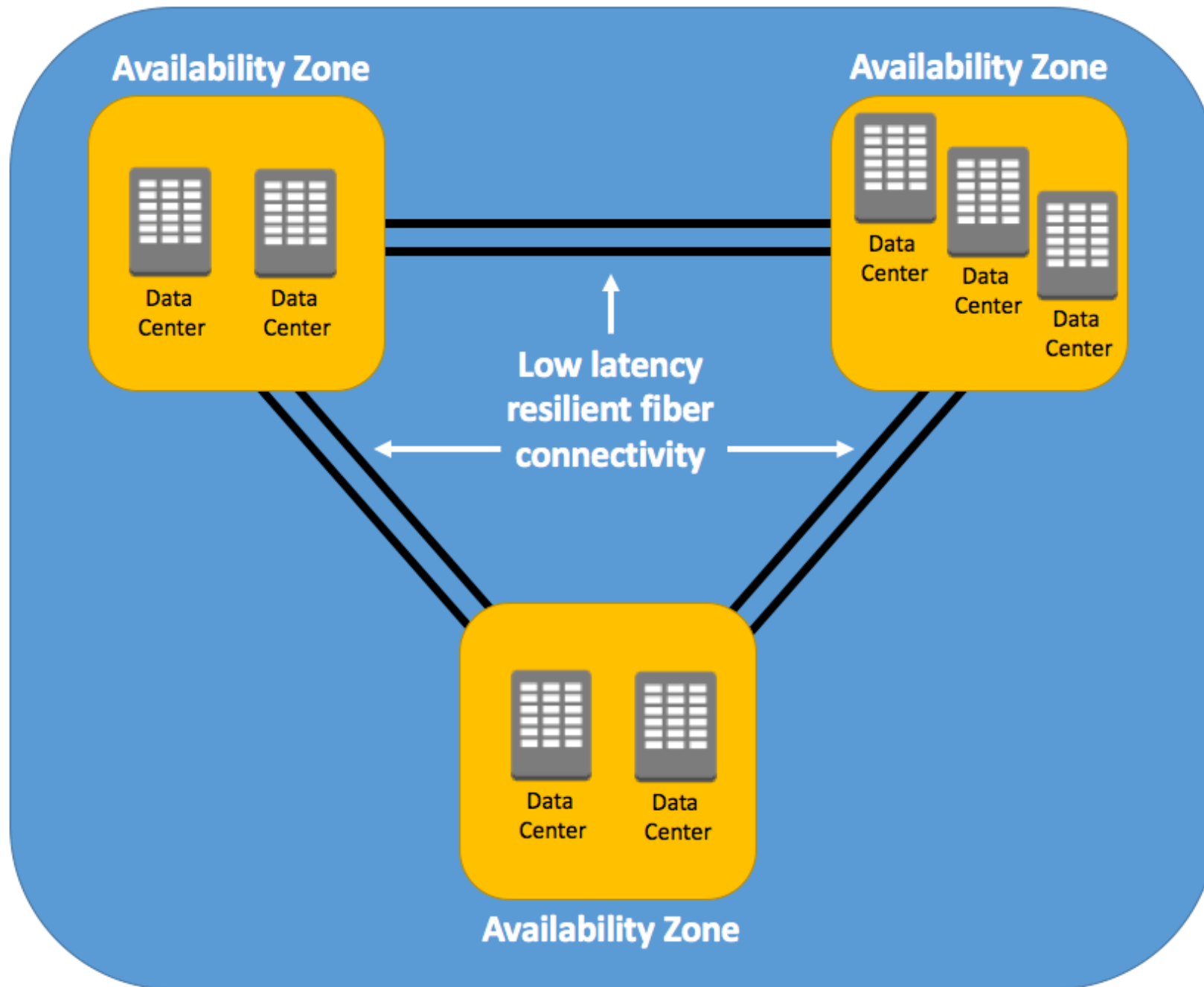
Global Infrastructure is designed and built to deliver a **flexible, reliable, scalable, and secure** cloud computing environment with high-quality **global network performance**.

- Collection of hardware and software resources that make up the cloud.
- Cloud providers maintain global data centers with thousands of IT infrastructure components like servers, physical storage devices, and networking equipment.

Benefits of Global Infrastructure

- Security
- Availability
- Performance
- Scalability
- Flexibility

REGION



Cloud Global Infrastructure

Data Center

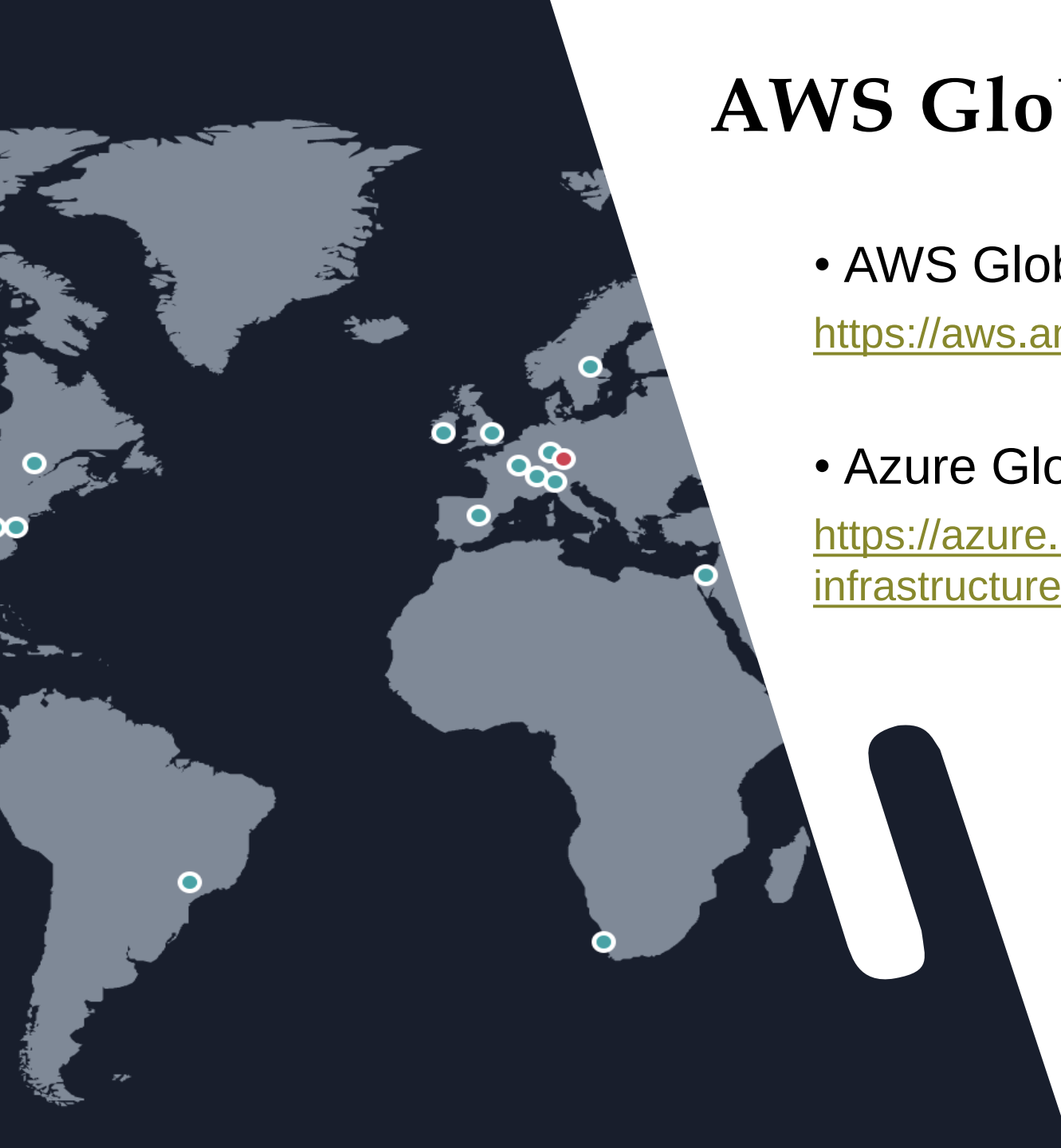


Availability Zone



Region





AWS Global Infrastructure

- AWS Global Infrastructure Map:

<https://aws.amazon.com/about-aws/global-infrastructure/#>

- Azure Global Infrastructure Map:

https://azure.microsoft.com/en-us/explore/global-infrastructure#tabs-pill-bar-oc9430_tab0

Selecting a Region

- Determine the right Region for your services, applications, and data based on these factors.
 - Data governance, legal requirements
 - Proximity to customers (latency)
 - Services available within the Region
 - Costs (vary by Region)

Availability Zones

- Each Region has multiple Availability Zones.
- Each Availability Zone is a fully isolated partition of the AWS infrastructure.
 - Availability Zones consist of discrete data centers
 - They are interconnected with other Availability Zones by using high-speed private networking
 - You choose your Availability Zones

Data Centers

- Data centers are where the data resides and data processing occurs.
- Each data center has redundant power, networking, and connectivity, and is housed in a separate facility.

AWS Services

Deployment & Management

Application Services



Amazon
SQS



Amazon
ElasticTranscoder



Amazon
SES



Amazon
AppStream



Amazon
CloudSearch

Mobile Services



Amazon
Cognito



Amazon
Mobile Analytics



Amazon
SNS

Enterprise Applications



Amazon
WorkDocs



Amazon
WorkSpaces



Amazon
WorkMail

Application Services

Administration & Security



AWS
DirectoryService



AWS
IAM



AWS
Trusted Advisor



AWS
Config



AWS
CloudTrail



Amazon
CloudWatch

Deployment & Management



Amazon
CloudFormation



AWS
OpsWorks



AWS
CodeDeploy

Analytics



Amazon
Kinesis



AWS
Data Pipeline



Amazon
EMR

Foundation Services

Compute



Amazon
EC2



AWS
Lambda

Storage & Content Delivery



Amazon
CloudFront



Amazon
Glacier



Amazon
S3



Amazon
EBS

Database



Amazon
DynamoDB



Amazon
RDS



Amazon
Redshift



Amazon
Elastic Cache

Networking



Amazon
Route 53



Amazon
VPC



AWS
Direct Connect

Azure Services

Platform Services

Media & CDN



Integration



Compute Services



Application Platform



Developer Services



Data



Intelligence



Analytics & IoT



How do you pay for Cloud Service Providers?

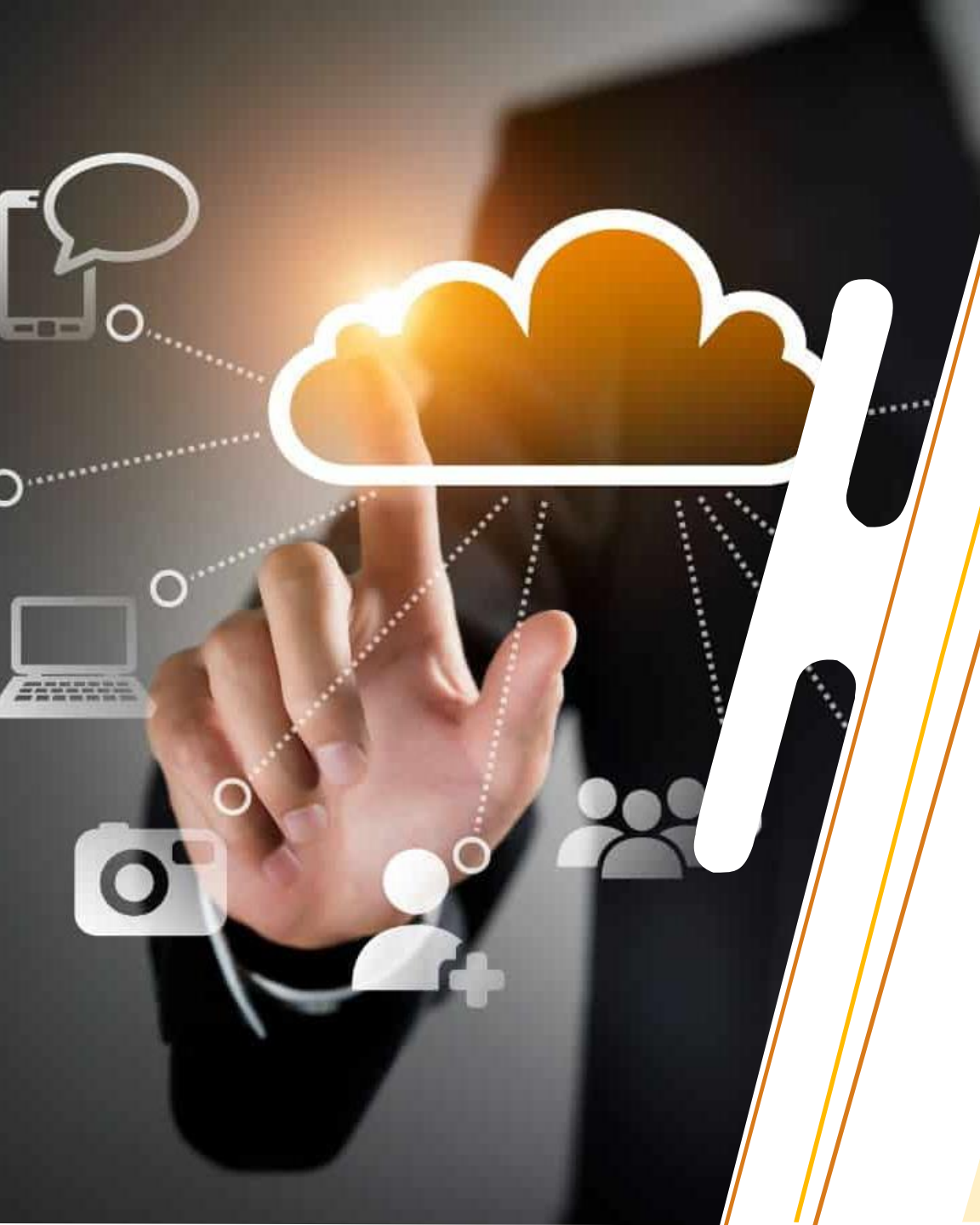
Pay for what you
use

Pay less when you
reserve

Pay less when you
use more

Custom Pricing

- Meet varying needs through custom pricing.
- Available for high-volume projects with unique requirements.



Next Session :
Cloud Services